SPL Project

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<u>Answer to the question-1 :</u>

```
#include<stdio.h>
#include<string.h>
struct Character
{
   char name[50];
   int age;
   char house[30];
   int wizard_points;
};
int main()
{
   struct Character c[5];
   int i;
   for(i = 0; i < 5; i++)
   {
      printf("Enter information for character %d\n", i + 1);
      printf("Enter name: ");
      scanf(" %[^\n]", c[i].name);
      printf("Enter age: ");
      scanf("%d", &c[i].age);
      printf("Enter house: ");
      scanf(" %[^\n]", c[i].house);
      printf("Enter wizard points: ");
      scanf("%d", &c[i].wizard_points);
      printf("\n");
   }
   for(i = 0; i < 5; i++)
   {
      printf("\nCharacter %d\n", i + 1);
      printf("Name: %s\n", c[i].name);
      printf("Age: %d\n", c[i].age);
      printf("House: %s\n", c[i].house);
      printf("Wizard Points: %d\n", c[i].wizard_points);
```

```
}
char search_name[50];
int found = 0;
printf("\nEnter character name to search: ");
scanf(" %[^\n]", search_name);
for(i = 0; i < 5; i++)
{
  if(strcmp(c[i].name, search_name) == 0)
  {
    found = 1;
    printf("\nCharacter Found!\n");
    printf("Name: %s\n", c[i].name);
    printf("Age: %d\n", c[i].age);
    printf("House: %s\n", c[i].house);
    printf("Wizard Points: %d\n", c[i].wizard_points);
  }
}
if(found == 0)
{
  printf("Character not found.\n");
}
int sum = 0;
for(i = 0; i < 5; i++)
{
  sum += c[i].age;
}
float average = (float)sum / 5;
printf("\nAverage Age: %.2f\n", average);
int longest = 0;
int shortest = 0;
for(i = 1; i < 5; i++)
{
  if(strlen(c[i].name) > strlen(c[longest].name))
  {
    longest = i;
  }
  if(strlen(c[i].name) < strlen(c[shortest].name))
  {
    shortest = i;
  }
}
printf("\nCharacter with Longest Name: %s\n", c[longest].name);
printf("Character with Shortest Name: %s\n", c[shortest].name);
return 0;
```

}

Line by Line Explanation :

a. #include<stdio.h>

→Used for input and output functions like printf and scanf.

b. #include<string.h>

→Used for string handling functions such as strcmp and strlen.

c. struct Character

→A structure is created to store character information together.

d.char name[50];

→Stores the character name.

e.int age;

→Stores the age of the character.

f.char house[30];

→Stores the Hogwarts house.

g.int wizard_points;

→Stores wizard points of the character.

h.struct Character c[5];

→Creates an array of 5 characters.

i.for(i = 0; i < 5; i++)

→Loop used to input data for all 5 characters.

j.scanf(" %[^\n]", c[i].name);

→Takes string input with spaces.

k.strcmp()

→Compares two strings for searching.

l.strlen()

→Finds the length of a string.

m.sum += c[i].age;

→Adds all ages for average calculation.

n.average = (float)sum / 5

→Calculates average age.

Longest and shortest name logic

→Uses string length comparison.

return 0;

Answer to the question-2 :

A structure is good for organizing character data because it can store different types of information together. In this program, each character has name, age, house, and wizard points. Using structure makes the program easy to write and understand.
I chose Harry Potter fiction because it is very popular and interesting. The house and wizard points are good characteristics for Harry Potter characters. House shows which Hogwarts house the character belongs to, like Gryffindor or Slytherin. Wizard points show the magical performance and achievements of the character.These characteristics match the Harry Potter story very well because houses and wizard activities are important parts of the fiction.